Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

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Declaration of competing interest

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Nguyễn Nhật Thanh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Ethics approval

Not applicable. The study involves no human participants or animal subjects; the real-data case uses a public, anonymised retail dataset.

Data and code availability

The synthetic demand data are generated by the code. The real-data case uses the public FreshRetailNet-50K dataset (Wang et al., 2025; CC BY 4.0), which is downloaded from Hugging Face as described in the replication package and processed by the code; no proprietary data are used. The code, configuration files, random seeds and result files that reproduce every table and figure are provided as supplementary material and will be made publicly available in a GitHub repository upon publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic), through the Claude Code assistant, to help write and restructure the simulation and experiment code, run the experiments, and draft and edit the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.